

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next Item →

1. Subtract the binary numbers 1010 from 1101. 1 / 1 point

0011

Feedback

Correct answer: 0011

Step 1: Subtract the numbers starting from the right-most digit:

$1 - 0 = 1$

$0 - 1 = 1$ (borrow one from the next left column; that column now becomes 0)

$0 - 0 = 0$

$1 - 1 = 0$

Step 2:

The final number is 0011.

2. Subtract the binary number 1001 from 1100. 1 / 1 point

0011

Feedback

Correct answer: 0011

Step 1: Subtract the numbers starting from the right-most digit:

$0 - 1 = 1$ (borrow 1 from the next left column)

$0 - 0 = 0$

$1 - 0 = 1$

$1 - 1 = 0$

Step 2:

The final number is 0011.

3. Subtract the binary number 10101 from 11011. 1 / 1 point

00110

Feedback

Correct answer: 00110

Step 1: Subtract the numbers starting from the right-most digit:

$1 - 1 = 0$

$1 - 0 = 1$

$0 - 1 = 1$ (borrow from the next left column)

$0 - 0 = 0$

$1 - 1 = 0$

Step 2:

The final number is 00110.

4. Multiply the binary numbers 10111 and 101. 1 / 1 point

1110011

Feedback

Correct answer: 1110011

Step 1: Perform binary multiplication similar to decimal multiplication:

$10111 \times 1 = 10111$

$10111 \times 0 = 00000$ (shifted one position to the left)

$10111 \times 1 = 10111$ (shifted two positions to the left)

Step 2: Add up the numbers:

$$\begin{array}{r} 10111 \\ 00000 \\ \underline{10111} \\ 1110011 \end{array}$$

5. Multiply the binary numbers 110 and 101. 1 / 1 point

11110

Feedback

Correct answer: 011110

Step 1: Perform binary multiplication similar to decimal multiplication:

$110 \times 1 = 110$

$110 \times 0 = 000$ (shifted left by 1 position)

$110 \times 1 = 110$ (shifted left by 2 positions)

Step 2: Add up the numbers:

$$\begin{array}{r} 110 \\ 000 \\ \underline{110} \\ 011110 \end{array}$$

6. Multiply the binary numbers 111 and 11. 1 / 1 point

10101

Feedback

Correct answer: 10101

Step 1: Perform binary multiplication similar to decimal multiplication:

$111 \times 1 = 111$

$111 \times 1 = 111$ (shifted one position to the left)

Step 2: Add up the numbers:

$$\begin{array}{r} 111 \\ \underline{111} \\ 10101 \end{array}$$

7. Add the binary numbers 1101 and 10111. 1 / 1 point

100100

Feedback

Correct answer: 0100100

Align the numbers and add:

$$\begin{array}{r} 01101 \\ \underline{10111} \\ 0100100 \end{array}$$

8. Add the binary numbers 110101 and 101110. 1 / 1 point

1100011

Feedback

Correct answer: 1100011

Align the numbers and add:

$$\begin{array}{r} 110101 \\ \underline{101110} \\ 1100011 \end{array}$$

9. Add the binary numbers 1011 and 1101. 1 / 1 point

10111

Feedback

Correct answer: 11000

Align the numbers and add:

$$\begin{array}{r} 01011 \\ \underline{01101} \\ 11000 \end{array}$$